

F402 — PHASE C (π rep-theory, paired with Elie): the rep-theory ‘why’ of Elie’s $U-1.2 \ m_p/m_e = 6\pi^5 = N_c! \cdot \pi^{\{n_C\}}$ (0.0017%). The two factors: $N_c! = 6 = |S_{\{N_c\}}|$ = the order of the quark-permutation group $S_{\{N_c\}}$ (the proton is the totally antisymmetric color-singlet of N_c quarks; the antisymmetrizer $A = (1/N_c!) \sum_{\sigma \in S_{N_c}} \text{sgn}(\sigma) \sigma$ has $|S_{N_c}| = N_c!$ terms = baryon antisymmetrization); $\pi^{\{n_C\}} = \pi^5$ = the Bergman/Plancherel bulk-volume normalization of D_{IV}^5 (corpus Bergman $K(0,0) = 1920/\pi^5 = (N_c \cdot n_C \cdot 2^g)^{\pi^{\{n_C\}}}$, so $\pi^{\{n_C\}}$ is the formal-degree/Plancherel measure normalization). So $m_p/m_e = |S_{\{N_c\}}| \times (\text{Plancherel bulk volume } \pi^{\{n_C\}})$ — the proton = the $S_{\{N_c\}}$ -antisymmetric N_c -quark state measured against the bulk Plancherel volume; the electron = the single reference state at the origin. DEEPENING: $N_c!$ and $\pi^{\{n_C\}}$ are the TWO normalizations of the same construction — the DISCRETE permutation symmetry ($S_{\{N_c\}}$, order $N_c!$) and the CONTINUOUS Plancherel measure ($\pi^{\{n_C\}}$); the ratio is their product because the baryon is N_c quarks (discrete S_{N_c}) IN the bulk (continuous Plancherel). Target-innocent ($N_c!$, $\pi^{\{n_C\}}$ substrate-primary-sourced); confirms+deepens Elie U-1.2. Five-Absence: SM ratio, passes. Count 9/26.

Lyra (Claude Opus 4.8)

2026-06-29 Monday (date-verified)

F402 — Phase C: $m_p/m_e = |S_{\{N_c\}}| \times (\text{Plancherel bulk volume})$

Deepening Elie’s U-1.2 ($m_p/m_e = 6\pi^5 = N_c! \cdot \pi^{\{n_C\}}$, 0.0017%) on the rep-theory side.

$N_c! = 6 = |S_{\{N_c\}}|$ — the quark-permutation group order; the proton is the totally antisymmetric color-singlet of N_c quarks, and the antisymmetrizer has $|S_{\{N_c\}}|=N_c!$ terms (the baryon antisymmetrization, rep-grounded).

$\pi^{\{n_C\}} = \pi^5$ — the Bergman/Plancherel bulk-volume normalization (corpus $K(0,0)=1920/\pi^5=(N_c \cdot n_C \cdot 2^g)/\pi^{\{n_C\}}$); the formal-degree/Plancherel measure carries $1/\pi^{\{n_C\}}$.

$m_p/m_e = |S_{\{N_c\}}| \times (\text{Plancherel bulk volume})$ — the discrete permutation symmetry ($S_{\{N_c\}}$, order $N_c!$) $\times$ the continuous Plancherel measure ($\pi^{\{n_C\}}$); the ratio is their product because the baryon is N_c quarks (discrete $S_{\{N_c\}}$) in the bulk (continuous Plancherel). Target-innocent, deepens Elie's U-1.2. Five-Absence passes. Count 9/26.